

PROJECT TESTING

1.Test Planning:

Objective

The objective of testing is to ensure that all features of EduGenie function correctly, reliably, and efficiently.

Testing Types

- Functional Testing
- Integration Testing
- UI Testing
- Performance Testing
- API Testing
- Browser Compatibility Testing

Tools Used

- Google Chrome
- Microsoft Edge
- FastAPI
- Uvicorn
- Postman (optional)

Expected Result

All AI modules should work correctly without crashes and generate accurate responses.

2. Test Cases

Test ID	Feature	Test Case	Expected Result	Status
TC-01	Question Answering	Ask a programming question	AI returns correct answer	Pass
TC-02	Concept Explanation	Explain Machine Learning	Simple explanation displayed	Pass
TC-03	Quiz Generator	Generate Java quiz	MCQ questions displayed	Pass

Test ID	Feature	Test Case	Expected Result	Status
TC-04	Learning Path	Enter Python	Learning roadmap generated	Pass
TC-05	PDF Upload	Upload educational PDF	Summary generated	Pass
TC-06	Voice Input	Speak a question	Speech converted to text	Pass
TC-07	Text-to-Speech	Click Speak	AI response read aloud	Pass
TC-08	Download PDF	Download response	PDF downloaded	Pass
TC-09	Download TXT	Download response	TXT downloaded	Pass
TC-10	Dark Mode	Toggle theme	Theme changes correctly	Pass

3.Test Results:

Functional Testing

- Question Answering – Passed
- Concept Explanation – Passed
- Quiz Generation – Passed
- PDF Summarization – Passed
- Learning Path – Passed

UI Testing

- Responsive Design – Passed
- Dark Mode – Passed
- Voice Input – Passed
- Copy Response – Passed

API Testing

- FastAPI Endpoints – Passed
- Gemini API Integration – Passed

Overall Result

All major functionalities were successfully tested and performed as expected.

4. Bug Report:

Issue 1

Problem

Gemini API quota exceeded.

Solution

Used a new API key and handled API exceptions.

Issue 2

Problem

Port 8000 already in use.

Solution

Stopped the existing Uvicorn process or used another port.

Issue 3

Problem

Speech Recognition unavailable in unsupported browsers.

Solution

Displayed an appropriate message to the user.

Issue 4

Problem

Invalid PDF upload.

Solution

Added validation before processing.

5. Performance_Testing

Response Time

- Question Answering – 2–5 seconds
- Quiz Generation – 3–6 seconds
- Concept Explanation – 2–5 seconds
- Learning Path – 2–4 seconds
- PDF Summary – 4–8 seconds

Browser Compatibility

Browser	Status
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Google Chrome	Supported
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Microsoft Edge	Supported
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Firefox	Supported
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Result

The application performs efficiently under normal usage conditions.